What Prices Cannot Tell a Market Supervisor: The Cost of Identifying Trader Groups and the Value of Class Disclosure

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Abstract

Prices in a market with feedback trading reveal the combined response of all trader groups, not the groups themselves. We characterize what public prices identify and what learning the rest costs. In a linear closed-loop market where each group tracks prices with exponential memory, prices identify only a merged response measure: groups with equal memories are inseparable however much data is collected, and separating K groups whose memories are $\Delta\eta$ apart takes of order $(\Delta\eta)^{-(4K-2)}$ experiments. That rate is the known mixture rate; we show that it survives the feedback loop, where the memories enter as zeros of the observed transfer function. The contributions are what the market setting adds. Passive observation of a running market is no cheaper than active probing, because the closed loop is minimum phase. For an observer who trades to learn, the sample requirements become executed quantity of order δ^{-2} and capital of order δ^{-4}. With a strategic insider and any causal dealer rule, economies with the same merged measure are outcome-equivalent, so what prices cannot identify is worth nothing to price-based traders; a dealer that misjudges the aggregate response pays or earns rents of first order in its error, which public data shrink at the ordinary rate. In an equilibrium with inventory-averse dealers, price impact and the persistence of mispricing are derived, and the equilibrium price is the same closed loop with effective parameters and unchanged memory rates, so every rate holds in equilibrium, also with a strategic insider. Positive feedback raises impact, and an explicit

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threshold in dealer risk aversion, inversely proportional to the feedback intensity, marks where liquidity disappears. Class-tagged flow removes the spacing penalty, and a supervisory example shows attribution changing the right action when prices are nearly identical. Several identification and Fisher-information results are machine-checked in Lean 4. Pre-registered tests on CFTC and Binance trader-class reports are mixed: 11 of 23 recorded outcomes pass; class reports constrain persistence far more tightly than prices but do not identify learning memory.

AI-use disclosure. Claude, ChatGPT/Codex and Gemini provided material assistance with research development, code, numerical checks, manuscript drafting and internal critique. ChatGPT/Codex additionally developed and checked Lean proofs in this revision. Human authors remain responsible for verifying the content and its scholarly contribution before submission; AI checks do not constitute independent review.

Keywords: algorithmic trading, identification, trader heterogeneity, market microstructure, system identification, minimax sample complexity, value of disclosure, audit trails.

JEL: C14, C32, C58, D83, G14, G18.

Code and data. All code, pre-specified test specifications and result files are public at https://github.com/Pratikn03/Fianince.

Reproducibility and verification. The appendix maps historical tables and figures to generating code and result files. Existing empirical results are retained with their original specifications, failures and limitations; this revision does not independently rerun every historical study. "Pre-specified" denotes the authors' dated specification and repository record, not independent registration. The new accounting study reproduces six archived response rows and records both the normal-quantile and conservative gates, including adverse configurations. The pinned Lean build and strict axiom audit verify typed declarations; the exact supervisory witness, conservative Gaussian event bound and acquisition algebra have explicit coverage below. Historical name mappings or declaration counts do not establish full manuscript equivalence. The stationary IV central limit argument, external approximation results, general-K supervisory construction and economic observation assumptions retain their stated limits. Code and receipts accompany this working paper; independent proof review remains a separate task.

1 Introduction

Markets are increasingly run by algorithms. Trend-following funds, market-making programs, execution algorithms and, more recently, AI trading agents all read recent prices, decide what to trade, and by trading move the prices they read. When something goes wrong, a sharp sell-off, a sudden loss of liquidity, a crowded trade unwinding, the first questions supervisors and risk managers ask are who was trading and how. Which group of traders drove the move? How quickly do they react, and how long do they remember what prices did? Could the same group do it again?

This paper asks a question that supervisors now face when they decide which trading records to require: which questions about who drives prices can public data answer, which can they answer only at a prohibitive cost, and what is it worth to be told the rest?

The question matters now. Trading is passing to learning algorithms and AI agents (Colliard et al., 2022; Dou et al., 2025), and regulators have begun to require firms to say which algorithm sent each order. Under MiFID II, EU trading venues must keep order records that identify the person or algorithm responsible for each investment and execution decision (European Commission, 2017; ESMA, 2016). The US Consolidated Audit Trail (CAT) is designed to record each order in US-listed stocks and options, and the account behind it, across exchanges and broker-dealers; its revised 2026 budget, published in March 2026, is about $147 million (SEC, 2012; CAT LLC, 2026). Such systems serve many purposes, among them enforcement, rebuilding market events after the fact, and surveillance. One question they raise, and the one we study, is how much of the information about who drives price moves could instead be learned from public data, and at what cost.

The question. A supervisor wants to know which group of traders drives prices. What can public prices tell it, what does it cost to learn the rest, and what are trader-class reports worth?

The answer.

1. *Prices see the crowd, not the groups.* They reveal one combined summary of all the groups. Two groups with the same memory look exactly like one larger group, however much price data is collected (Theorem 3.1).
2. *So supervision splits in two.* A market-wide measure depends on an aggregate response, subject to the observation and tail conditions. A targeted decision can require more information: two markets can look almost identical in prices while halting the largest group meets the prudential ceiling in one and not in the other (Proposition 7.4, machine-checked for two groups).
3. *Learning the groups from prices is very expensive.* The data needed grow as $(\Delta\eta)^{-(4K-2)}$ when the memories of K groups are $\Delta\eta$ apart: halving the gap multiplies the data by 64 for two groups and by 1,024 for three (Theorems 6.1 and 6.2). If memory is a general mixture, accuracy improves only with the logarithm of the data (Theorem S17.2).
4. *Class records change the information requirement.* When each class's flow is reported separately, the problem stays well conditioned however similar the groups are (Proposition 7.1). In 20 futures markets, the CFTC's weekly class report pins down the persistence of speculative positions 9 to 71 times more precisely than prices; a hypothetical common-root-T conversion gives about 1,350 at the median, but does not identify an equivalent sample requirement (Figure 1).
5. *What we do not show.* That fitted persistence is the traders' learning memory: the weekly reports constrain how persistent class positions are, which can also reflect turnover, reporting aggregation or inventory retention. Of 23 recorded outcomes, 11 failed (10 of the 19 primary ones; Table S29). One possible reason, which we have not tested, is that a reporting class mixes strategies that respond to prices with opposite signs.

We keep three statements apart throughout: what cannot be learned at all, what can be learned but only at high cost, and where estimates become unstable.

What is new and what is not. Table 1 separates the results we inherit from those the market setting adds. We claim novelty only for the rows marked new, and only for this model; Supplement S4 gives the closest results we know, with their statements.

Why prices hide the groups. The reason is the feedback loop. Algorithms trade on recent prices, their orders move prices, and those prices feed back into the algorithms. We

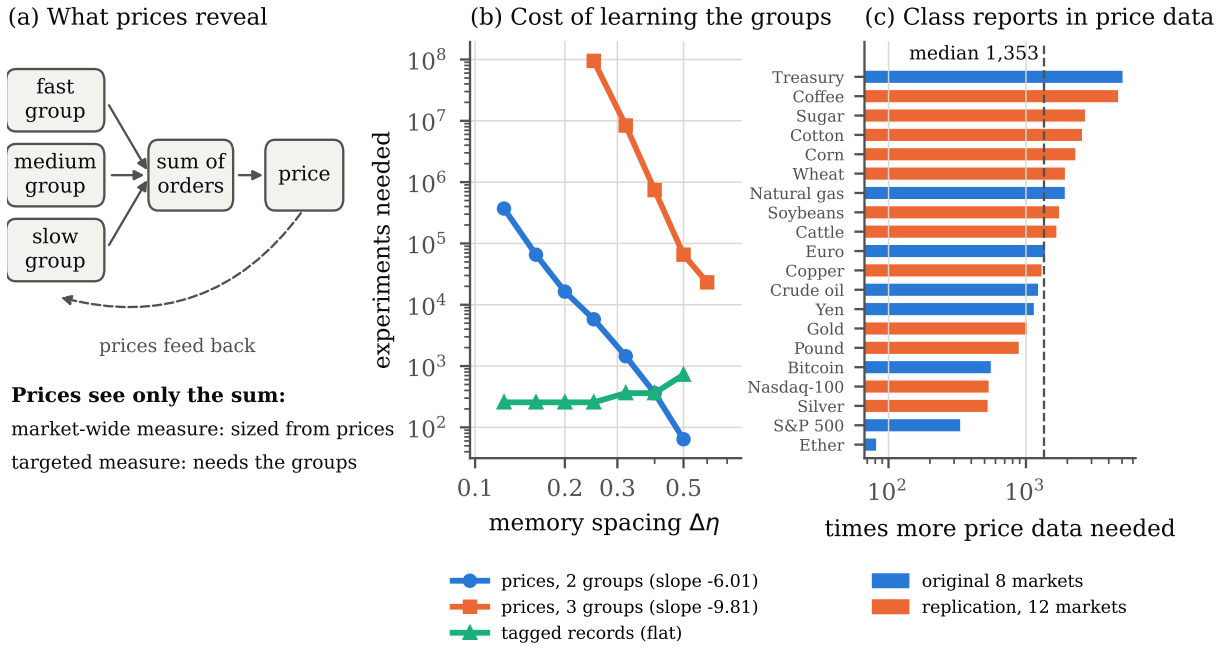


Figure 1: The paper in one figure. (a) Groups of traders reach the price only through the sum of their orders, so prices reveal the combined behaviour: enough to size a market-wide measure, not to decide whether halting one group is enough. (b) Pre-specified real-noise injection test (Supplement S16): experiments needed to recover the groups' weights against the spacing of their memories, with real one-minute Bitcoin price noise; price-only requirements grow with slopes −6.01 (two groups) and −9.81 (three groups; theory −6 and −10), tagged requirements stay flat. (c) Post hoc: how many times more data a price-only analysis would need to match the precision of the CFTC's weekly class report under standard root-T scaling, in the 20 markets of the registered tests (Table S26); the conversion assumes common root-T scaling and comparable estimands, which these data do not establish. Kind: registered results (b) and a calculation from registered results (c), `figs/make_fig_summary.py`.

model each group as tracking the price with a fading memory: recent prices count most, older prices count less, and one number, the group's memory rate, sets how fast the past fades. A desk behaves exactly this way if it uses a Kalman filter or a regularized online learning rule (Lemma 2.1). Because every group reaches the price only through the sum of all their trades, the price records the total and loses the parts. Think of several people pushing the same door: from the door's movement one can learn the total push and how it builds and fades, but two people who push in the same rhythm look like one stronger person.

Three ways to learn more. An observer that wants more than prices can wait and watch, which costs time and needs at least as many observations as an active prober needs experiments (Theorem 6.5); trade against the groups, which costs money, with capital rising

as $1/\delta^4$ (Theorem S24.2); or acquire labelled records, which costs whatever the reporting system costs but removes the explosion in data (Theorem 7.2). Supplement S26 gives the details.

How large are the costs? In a hypothetical benchmark (Supplement S23), an outside trader that re-learned two unknown algorithms each week with test trades would lose at least $15 million to $356 million a year per market. Regulators do not trade to learn; the figure prices the information that labelled records deliver without any trading.

Why this matters for finance. Supervisors learn which data to require for which question, exchanges learn which questions their account-level data already answer, risk managers get a realistic limit on what can be inferred about other traders, and researchers get one reason why models with several trader types are hard to estimate (Supplement S20).

What the data show. Two real reporting systems let us check the main prediction, that class records pin down what prices leave open. The U.S. Commodity Futures Trading Commission collects positions from large futures traders and publishes a weekly snapshot of them, aggregated by trader category. In a pre-specified test in eight futures markets (CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, crude oil and natural gas), these class positions pin down the persistence of the designated speculative class's positions to a range 9 to 71 times narrower than weekly prices allow, while prices alone leave it almost anywhere in the admissible range; a replication registered later on twelve new markets (metals, grains, softs, cattle, the Nasdaq-100 and the pound) holds in all twelve (Supplement S18). Converted into data, matching the precision of the weekly class report from prices alone would take, at the median market, about 1,350 times as much price data, the equivalent of thousands of years of weekly prices; this is hypothetical arithmetic, not a lower bound, because estimands and effective horizons differ and intervals reach parameter bounds. On Binance, in a pre-specified test on 30 markets never used before that matches the memory horizon of the two routes, the class data give the narrower interval in 29 of 29 evaluable markets, by a median factor of 7.3 (Supplement S19).

Passes and failures. Table S29 in the supplement lists all twenty-three recorded outcomes with their predictions: eleven passed or held, eleven failed and one secondary prediction was not supported. The primary CFTC result replicates on twelve new markets (12 of 12). The failures are part of the evidence: the CFTC's designated speculative class trend-followed in only four of the original eight markets (it did in eleven of the twelve new ones) and was the

less persistent class in only three of the new twelve; the reports read as weekly flows did not pin down how persistent that flow is; positions showed no distinct faster root that could be learning memory; Binance prices pinned down the combined response in only 5 of 30 markets, and the combined response was far cheaper to learn than the separate rates in only 10 of 30 in a longer follow-up; class data did not improve return forecasts; and in October 2026 prices did not forecast the top-trader position beyond its own persistence (1 of 30), and taker flow did not pin down the combined response (5 of 30), although its uncertainty shrank at the predicted root-T rate in all 30. Supplement S21 reports the robustness checks.

Two supplementary results, on markets that share eigenvalues and price variance but not short-run risk and on how reduced-form feedback is measured, are summarized in Supplement S22.

How to read the paper. Every main result is followed by a short paragraph headed "In words"; Supplement S30 gives a reading guide and the main symbols in plain words.

What we do not claim. The identification results hold for the stated linear model and for the stated observation contracts, that is, for what the observer is assumed to see. Sections 5–7 treat price impact and the persistence of pricing errors as known primitives. Section 4 derives them in a dealer equilibrium, in which the price follows the same closed loop with effective parameters, so the results hold there (Theorem 4.2); that equilibrium has one dealer sector and, in Theorem 4.6, a strategic insider with short-lived information only. The model also assumes the observer knows the fundamental value. Real data meet neither assumption. They are not general limits on surveillance with participant-level records, which can be far more informative (Propositions S24.3 and 7.1). The matching upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are reduced-form diagnostics; they do not identify structural learning parameters. Our pre-specified real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Supplement S19). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 7.1, not an instance of it. We hold the algorithms' memories and intensities fixed across disclosure regimes; disclosure that reveals positions may lead informed desks to trade less or to split orders across accounts, which we do not model.

1.1 Relation to the literature

Three literatures meet here. In finance, learning algorithms and AI agents in markets are studied by simulating or modelling them (Colliard et al., 2022; Dou et al., 2025); traders who differ in horizon or speed are studied in equilibrium (Cespa and Vives, 2015; Biais et al., 2015; Crouzet et al., 2020); the exponentially fading beliefs we use are the extrapolative beliefs of Barberis et al. (2015, 2018); and heterogeneous-agent models estimated from prices often leave the type-separating parameters imprecise (Brock and Hommes, 1998; Amilon, 2008; Kukacka and Barunik, 2017), for which our results give one reason. Price formation follows the linear-impact tradition (Kyle, 1985; Glosten and Milgrom, 1985; Hasbrouck, 1991) and the transient-impact literature (Obizhaeva and Wang, 2013; Gatheral, 2010; Alfonsi et al., 2012); trader types have been inferred from public data, as in the probability of informed trading (Easley et al., 1996), and from account data (Kirilenko et al., 2017). Learning by trading has an equilibrium counterpart in price experimentation (Leach and Madhavan, 1993), and execution that learns impact while trading is studied by Moallemi et al. (2008) and Park and Van Roy (2015). Technically, the closest work is closed-loop identification (Ljung, 1999; Forssell and Ljung, 1999), weak identification (Andrews and Cheng, 2012), and super-resolution and mixture estimation (Moitra, 2015; Heinrich and Kahn, 2018; Wu and Yang, 2020; Batenkov et al., 2021; Katz et al., 2023). We do not claim the K-group exponent as new: in mixture language it is the square of the condition number for K clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and Prony's method attains it (Katz et al., 2023); the δ^{-4} law for one against two close components is classical in resolution analysis (Shahram and Milanfar, 2005); logarithmic rates are known for mixtures of exponentials (Rebafka and Roueff, 2015); and labelled samples are known to be more informative than unlabelled ones (Castelli and Cover, 1996). The model-specific contribution proposed here is the closed-loop form (the memories are zeros, not poles, of the observed transfer function, so they must first be extracted from the loop), the passive equivalence, the translation of sample sizes into executed quantity, capital and money, and the value of class records, including a decision in which attribution changes the right action. Supplement S4 gives the full comparison, with tables of the closest results.

1.2 Outline

Section 2 sets out the model and Section 3 the identification boundary. Section 4 derives price impact and persistence from inventory-averse dealers and shows that the boundary is an equilibrium object. Section 5 turns memory resolution into executed quantity, capital and money for a known pair, and Section 6 treats unknown mixtures, general memory and

passive data. Section 7 states what disclosure buys and when it changes a supervisor's decision, and tests it on the CFTC's mandated trader-class reports and on Binance's class ratios. Section 8 draws the implications for supervision and Section 9 concludes. The online supplement contains the proofs, the extended literature review and the map of results, the discussion of the assumptions, general memory, the real-noise test, the trader-class tests in full, further results and robustness checks, the hypothetical dollar illustration, a finite-horizon decision example, the spectral blind spot, the measurement of reduced-form feedback, the formal-verification statement and the reproducibility map.

2 Model and motivating examples

The model in words. The market has one price and K groups of traders. Each period three things happen. First, the price moves: it drifts back toward its fundamental value, and it is pushed by the groups' trades, by any test trade and by news. Second, each group updates its view by mixing the newest price into its running average, with weight η_i on the newest price and $1-\eta_i$ on its old view. Third, each group trades in proportion to its view, with intensity q_i. A group with a large η_i has a short memory and reacts quickly, like a short-horizon trend follower; a group with a small η_i has a long memory and reacts slowly. The equations below write this down.

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1-\eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0,1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1-\eta_i$ for the memory rate and $w_i = q_i\eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j\geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z-a)D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1-S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = Jz_{t-1} + bv_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\,(1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma 2.1 (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2 \sigma_\nu^2}}{2\sigma_\nu^2} \in (0, 1). \tag{7}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. A tabular value update with learning rate η at a fixed state has the same form; general reinforcement-learning policies need not.*

Remark 2.2 (Why exponential memory). Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced-form belief state $y_{i,t}$. When desk i chooses order flow $u_{i,t}$ subject to $I_{i,t} = (1 - \delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and an inventory cap $|I_{i,t}| \leq \bar{I}_i$ to maximize expected continuation return $\beta_i y_{i,t} u_{i,t}$ net of quadratic execution cost $\frac{1}{2}\lambda_{\text{tc},i} u_{i,t}^2$ and quadratic inventory holding cost $\frac{1}{2}\gamma_{\text{inv},i} I_{i,t}^2$, its interior first-order condition gives $u_{i,t}^* = q_i y_{i,t} - \psi_{\text{inv},i} I_{i,t-1}$ with market-response weight $q_i = \beta_i/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$ and inventory mean-reversion $\psi_{\text{inv},i} = \gamma_{\text{inv},i}(1 - \delta_{\text{inv}})/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$,

which shrinks toward zero as trading costs or inventory penalties tighten and destabilizes the market when the closed-loop spectral radius reaches one. This filtering-and-control recursion specifies an individual agent's decision rule, not a complete economic equilibrium (dealer–trader equilibrium is analysed in the companion paper).

Two further simplifications matter for reading the results: the observer is taken to know the fundamental value p^*, so the pricing error x_t is observed, and the dealer sector is a fixed linear impact κ rather than an equilibrium quote. The paragraph before Supplement S18 states what the data tests lose when neither holds. Assuming that the observer knows p^* gives it more information than any real observer has, so the lower bounds hold a fortiori when p^* is unknown; Corollary 6.3 proves that baseline demeaning eliminates unknown p^* without affecting the upper-bound rate.

The model applies these rules to the pricing error x_t: a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a signal of a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). With $q_i > 0$ such a desk buys after rises, as trend followers do, and nothing in the model says this is profitable; the lemma shows only that the update has the exponential form. In these two illustrative families of algorithms the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio or its regularization. That is the kind of private information a supervisor would like to recover, although real trading algorithms need not belong to either family. Section 3 shows why prices reveal it only in merged form.

Lemma 2.3 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_{\ell} \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_{\ell} \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

As an illustration of the lemma (not part of its proof), $(-1)^n \Delta^n h_j \geq 0$ for $n \leq 4$ held in all 892 stable random positive economies tested. Lemma 2.3 matters twice: it makes probing costs positive (Section 5), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. The companion paper locates amplification in order-book depth instead.

Assumptions. Supplement S14 discusses the six modelling choices behind the results: linearity, price-first timing within a period, exponential memory (relaxed in Supplement S17),

algorithms that react to being probed, known primitives, and the observation contracts. Unknown primitives can only make separation harder, so the lower bounds remain valid; Corollary 6.3 proves that estimating (p^*, κ, a) at the parametric rate leaves the $\Theta((\Delta\eta)^{-(4K-2)})$ upper-bound rate unchanged.

2.1 The observer, its options and its costs

The results are stated from the side of one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. Its *target* is attribution, recovering the merged weight and memory rate of each group to a stated accuracy, or deciding between two candidate populations at a stated error. Before it observes, it knows the price persistence a, the impact κ and the law of the observation noise, and, for the K-group results, the number of groups; it does not know their rates or weights. These are the conditions under which the lower bounds are statements about the data rather than about the observer's ignorance; unknown primitives can only enlarge the set of candidate populations.

The observer has three options, and each result prices one of them.

(a) *Wait.* Record stationary prices driven by natural innovations, without trading. The cost is time: the number of observations needed (Theorems 6.4 and 6.5).

(b) *Probe.* Execute a sequence v_t of trades against the groups and record the price response. Probes execute at the average price of a linear book after the uncontrolled state update; the prober bears the execution loss $\frac{1}{2}v^\top T_h v$ plus fees (Lemma S24.1) and must fund margin throughout. The counted resource is the number of independent experiments, each starting from the reset state, or the executed quantity and capital of a single experiment (Theorems 5.1–S9.2).

(c) *Acquire disclosure.* Receive each class's flow or position, observed with noise, at an operating cost per supervisory cycle and a compliance cost per market (Proposition 7.1, Theorem 7.2, Proposition S27.1).

Within each period the order is: the probe (if any) executes and the price updates, the groups update their states on the new price, and the observer records the price with noise. The groups do not change their rules in response to being observed; Supplement S14 discusses what happens if they do. Probing experiments start from the reset state. A probe sent into a running market adds the free response of the current state, which by linearity superposes with the forced response; if the state is unknown it acts as a nuisance term, which can only make separation harder, so the lower bounds still hold, while the upper bounds as stated need the reset state or a known one. Independent resets are a benchmark experiment that no

outside participant can run in a live market; we use them because they make the information content of the data exact, and the economic comparisons say whenever they rely on them. A numerical check without resets, in which a known random probe is sent into one long running market whose groups' state is unknown, keeps the $(\Delta\eta)^{-6}$ order for $K = 2$ and raises the constant by a factor of about 1.4 to 6 (Supplement S3.8). Regulators do not, in practice, trade to learn; the probing option describes an outside participant, or prices the information a supervisor would otherwise have to buy by trading.

Three money quantities appear and are kept apart: the *capital required*, the equity an account must hold so that its margin constraint is never breached while it probes (Theorem S24.2 and Lemma S24.4); the *round-trip trading loss*, the execution loss of a probe that ends flat, which is what the account actually spends (Theorems S9.2, S9.3 and S9.4); and the *annualised loss* of renewing that probe weekly, used for the comparison with audit-trail budgets (Supplement S8). The lower bounds hold for the stated probe classes, deterministic probes and, in Theorem S9.3, non-anticipating sequential probes; they are not lower bounds on information obtained through other channels, which is the point of Section 7. Two rates recur and measure different things: δ^{-4} is the capital (and the average-power horizon) for a *known* symmetric pair whose rates differ by 2δ, while $(\Delta\eta)^{-(4K-2)}$, which is $(\Delta\eta)^{-6}$ for $K = 2$, is the number of experiments for K *unknown* groups with spacing $\Delta\eta$.

3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

Theorem 3.1 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1]. \tag{8}$$

Write $a' = a + \kappa M(\{0\})$, so that cohorts with $\eta = 1$ (rate 0) act as part of the anchor. For $a' \neq 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M) \cap (0, 1)|$, counting distinct positive rates with nonzero merged weight after signed cancellation; for $a' = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M) \cap (0, 1)|$. A cohort with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with all rates in $[\epsilon, 1-\epsilon]$ for some $\epsilon > 0$. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\ge\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{C_{\mathcal{F}}^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{9}$$

where $C_{\mathcal{F}}$ depends only on the family, through ϵ, the bound on $|H|$ implied by uniform stability, and the total weight q.

In words. If two trading desks update on prices with the same memory, no amount of price data can tell them apart or say how much each trades: prices reveal only how much trading happens at each memory speed. Desks with different memories can be told apart in principle, and the rest of the paper prices how hard that is.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence we verify in Section 6: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

Proposition 3.3 (Conditional endogenous-pricing invariance)**.** *Let $\kappa = \mathcal{K}(\Gamma_M(\kappa))$ on a common admissible interval, with identical innovation laws, initial-state conventions and other primitives across economies. Here $\Gamma_M(\kappa)$ is the specified covariance record at a fixed impact and $\mathcal{K}$ is a maintained dealer rule, not a derived general equilibrium.*

(i) If equal merged measures imply the same map Γ_M and the fixed point is unique (or a common selection rule depends only on that map), equal merged measures give equal selected impacts.

(ii) Suppose $\mathcal{K}$ is L_K-Lipschitz, each covariance map is L_Γ-Lipschitz in impact, $L_K L_\Gamma < 1$,

and $\sup_\kappa \|\Gamma_A(\kappa) - \Gamma_B(\kappa)\| \le \epsilon$. *For existing fixed points,*

$$|\kappa_A - \kappa_B| \le \frac{L_K \epsilon}{1 - L_K L_\Gamma}. \tag{10}$$

Thus a covariance discrepancy of order $(\Delta\eta)^{2K-1}$ *preserves that order only under these uniform conditions.*

Limits of the pricing extension. Anonymous flow alone implies neither a covariance-only dealer objective nor uniqueness or continuity of its equilibrium. Multiple equilibria, threshold quote changes, private information and inventory-dependent rules are outside this proposition; for the question of invariance, Theorem 4.5 below covers them. Preserving a price-record testing bound additionally requires a uniformly Lipschitz observation law with the same nondegenerate noise; adding anonymous flow is a different statistical experiment and requires its own divergence bound. We therefore make no unconditional claim that endogenous pricing or additional aggregate-flow observations preserve the price-only lower bound.

4 Equilibrium foundations

The identification results take the price impact κ and the persistence a of pricing errors as given, and nobody in the model optimizes. This section derives both from dealers who dislike holding inventory. It then shows four things:

- the equilibrium price is the paper's closed loop with effective parameters, so the identification boundary and the rates hold in equilibrium, and positive feedback creates an explicit fragility threshold;
- a strategic insider cannot change it;
- misjudging the aggregate response creates rents of first order;
- targeted actions still need group-level information, and class-tagged disclosure keeps its value in equilibrium.
- a strategic insider with short-lived information changes neither conclusion, while positive feedback makes prices less informative about her news.

An inventory equilibrium.

- *Fundamental and price.* The fundamental F_t is a public random walk with innovation variance σ_F^2, and the price is $p_t = F_t + x_t$.

- *Cohorts.* They trade $u_{i,t+1} = r_i u_{i,t} + w_i x_t$, the paper's exponential memory with $w_i = q_i \eta_i$ and $r_i = 1 - \eta_i$.
- *Exogenous flow.* $\xi_t \sim N(0, \sigma_\xi^2)$, i.i.d., stands for noise trades or the observer's probe. Customer flow is $y_t = \sum_i u_{i,t} + \xi_t$.
- *Dealers.* Competitive dealers absorb the flow and lay off a share $1-\bar{a}$ of their inventory each period, so $H_t = \bar{a} H_{t-1} - y_t$. Each overlapping generation of dealers has CARA risk aversion A and holds H_t to the next date: $H_t = (E_t p_{t+1} - p_t)/(AV)$, with $V = \mathrm{Var}_t(p_{t+1})$.

We study stationary equilibria in which x_t is linear in the dealers' inventory and the cohorts' next trades. Write $G(D) = \sum_i w_i/(D - r_i) = \int dM(r)/(D - r)$ for the Stieltjes transform of the merged measure.

Theorem 4.1 (Price impact and persistence in an inventory equilibrium)**.** *Consider stationary linear equilibria of the model above.*

(i) Characterization. *The price law*

$$x_t = -\kappa H_t + \kappa \sum_i \frac{u_{i,t+1}}{D - r_i}$$

with $D \neq 0$ and $D \neq r_i$, is an equilibrium exactly when the induced dynamics are stable and

$$D = 1 - \kappa\, G(D), \qquad \kappa(D - \bar{a}) = A\, V\, D, \qquad V = \sigma_F^2 + \frac{\kappa^2 \sigma_\xi^2}{D^2}. \tag{11}$$

An unexpected unit of flow moves the price by κ/D.

(ii) No feedback. *Then $D = 1$ and the price law is exactly the paper's $x_t = \bar{a}\, x_{t-1} + \kappa\, y_t$. The persistence of pricing errors is the dealers' carry share, $a = \bar{a}$, and the impact κ solves $\kappa(1-\bar{a}) = A(\sigma_F^2 + \kappa^2 \sigma_\xi^2)$. An equilibrium exists if and only if $\Delta = (1-\bar{a})^2 - 4A^2\sigma_F^2\sigma_\xi^2 \geq 0$. When $\Delta > 0$ there are two, $\kappa_\pm = \big((1-\bar{a}) \pm \sqrt{\Delta}\big)/(2A\sigma_\xi^2)$; the low-impact one $\kappa_- \approx A\sigma_F^2/(1-\bar{a})$ is the price of dealer inventory risk.*

(iii) Invariance. *The coefficients (κ, D, V), and hence the price process, depend on the cohorts only through G, and so only through the merged measure M. This is Theorem 4.5 in equilibrium form.*

(iv) Small feedback. *Let $\Delta > 0$, fix $\delta > 0$ with $\max_i r_i < 1 - \delta$, and let $\epsilon = \sup_{|D-1| \leq \delta}(|G(D)| + |G'(D)|)$ be small. Then there is a unique equilibrium near*

$(\kappa_-, 1)$. *With* $G(1) = \sum_i q_i$, *the aggregate feedback intensity,*

$$D = 1 - \kappa_- G(1) + o(\epsilon), \qquad \kappa = \kappa_- + \frac{\kappa_-^2(\bar{a} + 2A\kappa_-\sigma_\xi^2)}{\sqrt{\Delta}}\, G(1) + o(\epsilon).$$

Positive feedback raises the impact κ, *and it raises the impact of an unexpected flow shock,* κ/D, *by more.*

In words. Dealers set the price impact and the persistence of mispricing. Impact is what they charge for carrying inventory risk; persistence is how slowly they lay that inventory off. Without feedback traders, this equilibrium is exactly the closed loop the paper starts from. Feedback traders enter only through their merged measure, as Theorem 4.5 requires. Positive feedback raises price impact, and dealers price their predictable future trades in advance (the terms in $u_{i,t+1}$). Feedback trading also makes liquidity fragile, not only more expensive: beyond a threshold in dealer risk aversion that falls with the feedback intensity, no equilibrium exists (Theorem 4.4).

Machine check. `Paper1InventoryEquilibrium.lean` checks the algebraic core of part (i):

- the dealers' first-order condition holds in every state;
- the next price loads on the next flow shock with coefficient κ/D;
- the no-feedback existence condition.

Part (iii) is proved on paper. For part (iv), `Paper1InventoryIFT.lean` applies Mathlib's inverse function theorem along a one-parameter family of feedback $G = \epsilon g$, with g strictly differentiable at $D = 1$. For all small ϵ it gives a zero near $(\kappa_-, 1)$, its uniqueness in a fixed neighbourhood, and the two first-order derivatives in the statement (`p1_inventory_small_feedback`). The version that is uniform over all small cohort measures, in the norm of the statement, is proved on paper.

The identification results hold in equilibrium. With feedback, dealers price the cohorts' predictable trades in advance, so the response of prices to flow is no longer the reduced form of Section 2 with the primitive (a, κ). The next theorem shows that it is still that closed loop, exactly and path by path, with new parameters and the same memory rates.

Theorem 4.2 (The equilibrium is the paper's closed loop)**.** *Take an equilibrium of Theo-*

rem 4.1 with $D \neq \bar{a}$, and set

$$a^* = \frac{\bar{a}}{D}, \qquad \kappa^* = \frac{\kappa}{D}, \qquad \tilde{w}_i = \frac{D - \bar{a}}{D - r_i}\, w_i.$$

(i) Reduced form, pathwise. *Let $\tilde{u}_{i,t} = \frac{D-\bar{a}}{D-r_i} u_{i,t}$. For every realization, every initial state and every t,*

$$x_t = a^* x_{t-1} + \kappa^* \Big(\xi_t + \sum_i \tilde{u}_{i,t} \Big), \qquad \tilde{u}_{i,t+1} = r_i \tilde{u}_{i,t} + \tilde{w}_i x_t.$$

So the price follows the closed loop (1)–(2) with persistence a^*, impact κ^*, exogenous flow ξ in place of the probe, and cohorts with the same memory rates and merged weights $\tilde{w}_i$. Its transfer function from flow to price is (4) with $(a^*, \kappa^*, \tilde{w})$. The equilibrium is stable if and only if this closed loop is.

(ii) Zeros. *For $\bar{a} > 0$, distinct rates in $(0, 1)$ and nonzero weights, the memory rates are the transmission zeros of the equilibrium price response, and none cancels (Theorem 6.4(i) with the effective parameters).*

(iii) Boundary. *The effective merged measure is $\tilde{M} = h\, M$, where $h(r) = (D - \bar{a})/(D - r)$ is positive. The effective parameters determine the equilibrium ones: $D = 1 - \kappa^* \tilde{M}([0, 1])/(1 - a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. So the merged measure remains exactly what equilibrium prices identify (Theorem 3.1).*

(iv) The rates survive. *Let $(a^*, \kappa^*, W, r_c, \Delta\eta)$ satisfy the hypotheses of Theorem 6.1 with $(a, \kappa) = (a^*, \kappa^*)$. Then the two K-cohort economies of that theorem are the price responses of two inventory equilibria with identical primitives $(A, \sigma_F^2, \sigma_\xi^2, \bar{a})$ and identical (κ, D):*

- *$D = 1 - \kappa^* W/(1 - a^*)$, which exceeds every rate;*
- *$\bar{a} = a^* D$ and $\kappa = \kappa^* D$;*
- *any $\sigma_F^2, \sigma_\xi^2 > 0$, with $A = \kappa^* D(1 - a^*)/(\sigma_F^2 + \kappa^{*2} \sigma_\xi^2)$;*
- *cohort weights $w_i = \tilde{w}_i (D - r_i)/(D - \bar{a}) > 0$.*

Hence an observer of equilibrium prices who knows the primitives needs $N \geq c\, \sigma^2 (\Delta\eta)^{-(4K-2)}$ experiments. Theorem 6.2 applies to the effective parameters, and the true weights follow from the effective ones by a Lipschitz map. So $N^* \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$ in equilibrium.

In words. Dealers' anticipation does not hide the cohorts and does not move them. It replaces the persistence and impact by $\bar{a}/D$ and κ/D, and it rescales each cohort's weight by $(D-\bar{a})/(D-r_i)$. Because that factor rises with r_i, slow-memory cohorts weigh more in equilibrium prices. The memory rates are untouched.

- Every result of Sections 3, 5 and 6 that concerns the map from order flow to prices therefore holds in equilibrium: identification, sample complexity, passive observation, and probing costs computed from the price path. Its constants are evaluated at the effective parameters.
- The lower-bound pair is not an artefact of the reduced form. Two economies with the same dealers, the same risk aversion and the same impact differ only in their cohorts, and they are as hard to tell apart as Theorem 6.1 says.
- Part (i), the realization in part (iv) and the effective cohort recursion are machine-checked (`Paper1EquilibriumReducedForm.lean`).
- One assumption is new: the observer's probe is part of the exogenous flow that dealers price as noise. Dealers who recognized a probe would price it differently.

Corollary 4.3 (Disclosure in equilibrium)**.** *Take a positive-impact equilibrium of Theorem 4.1 with positive weights and $D > \max_i r_i$.*

(i) The cohorts' flows obey $u_{i,t+1} = r_i u_{i,t} + w_i x_t$ with their true *weights. The price is the stable closed loop of Theorem 4.2, driven by ξ, with positive effective weights. So Proposition 7.1 applies: tagged flow identifies each cohort's (r_i, w_i), with a limiting covariance that has no singularity as rates collide.*

(ii) On classes in which both the true and the effective weights lie in compact positive intervals, Theorem 7.2 holds in equilibrium. Tagging cuts the experiments needed by a factor of order $(\Delta\eta)^{-(4K-2)}$ for weights and $(\Delta\eta)^{-(4K-4)}$ for rates.

In words. What makes disclosure valuable survives the dealers' optimization. Prices still hide the groups at the same polynomial cost, and each group's own flow still reveals that group's memory and weight directly.

Theorem 4.4 (Fragility: where liquidity disappears)**.** *Let there be one feedback cohort with weight $w > 0$ and memory rate $r \in (0,1)$, with $\bar{a} \in [0,1)$ and $\sigma_F, \sigma_\xi > 0$. Call an equilibrium of Theorem 4.1 with $\kappa > 0$ and $D > 0$ a positive-impact equilibrium. For $D \in (m,1)$ with $m = \max(\bar{a}, r)$, let*

$$\kappa_w(D) = \frac{(1-D)(D-r)}{w}, \qquad \Psi_w(D) = \frac{\kappa_w(D)\,(D-\bar{a})\,D}{\sigma_F^2 D^2 + \kappa_w(D)^2 \sigma_\xi^2}, \qquad A^*(w) = \max_{D\in(m,1)} \Psi_w(D).$$

(i) Threshold.

- *The positive-impact equilibria are exactly the pairs* $(\kappa_w(D), D)$ *with* $D \in (m, 1)$ *and* $\Psi_w(D) = A$*, and every one of them is stable.*
- *A positive-impact equilibrium exists if and only if* $A \le A^*(w)$*.*
- *For* $A < A^*(w)$ *there are at least two; for* $A > A^*(w)$ *there are none.*

(ii) Monotonicity. $A^*(w)$ *is strictly decreasing in* w*. As* $w \downarrow 0$ *it tends to* $A_0^* = (1 - \bar{a})/(2\sigma_F\sigma_\xi)$*, the threshold of Theorem 4.1(ii).*

(iii) Bounds and rate. *For every* $w > 0$*,*

$$A^*(w) < \min\left\{A_0^*,\ \frac{\Phi^*}{\sigma_F^2\, w}\right\},$$

where

$$\Phi^* = \max_{D\in[m,1]} \frac{(1-D)(D-r)(D-\bar{a})}{D} \le \frac{(1-\bar{a})(1-r)^2}{4},$$

and $w\,A^*(w) \to \Phi^*/\sigma_F^2$ *as* $w \to \infty$*. In terms of the feedback intensity* $G(1) = w/(1 - r)$*, strong feedback gives* $A^* \approx \Phi^*/(\sigma_F^2(1 - r)G(1))$*, and always* $A^* < (1 - \bar{a})(1 - r)/(4\sigma_F^2 G(1))$*.*

(iv) Several cohorts. *Let all weights be positive.*

- *The positive-impact equilibria with* $D > \max_i r_i$ *are exactly the pairs* $((1 - D)/G(D), D)$ *with* $D \in (\max(\bar{a}, \max_i r_i), 1)$ *at which the analogue of* Ψ_w*, with* $\kappa = (1 - D)/G(D)$*, equals* A*.*
- *Every such equilibrium has*

$$A < \min\left\{A_0^*,\ \frac{(1-\sqrt{\bar{a}})^2}{\sigma_F^2\, G(1)}\right\},$$

  whatever the number of cohorts and their rates.
- *Every such equilibrium is stable.*

In words. With positive feedback, dealers can make a market only if their dislike of risk is below a threshold, and the threshold falls as feedback grows: for strong feedback it is inversely proportional to the feedback intensity. Past the threshold, no price impact is consistent with dealers' optimization. Below it, every equilibrium has stable prices. Liquidity does not erode through growing instability; it disappears when the two equilibria merge, at a point that stable prices give no warning of. At the calibration of Supplement S28 ($\bar{a} = 0.6$, $r = 0.7$,

$\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$), A^* is 0.40 without feedback, 0.25 at $G(1) = \frac{1}{15}$ and 0.034 at $G(1) = \frac{2}{3}$, and $w\,A^*$ approaches $\Phi^* = 0.00688$. The necessary conditions behind parts (i) and (iii) are machine-checked (`Paper1EquilibriumReducedForm.lean`). With several cohorts, part (iv) gives a threshold that depends on the population only through its total feedback intensity $G(1)$. At the same calibration, numerically, the exact thresholds for two and three cohorts lie between 28% and 86% of that bound, and the equilibria on the threshold curve are stable. Every such equilibrium is stable (part (iv)), as the scan also shows.

Theorem 4.5 (Equilibrium invariance)**.** *Consider a market with the following participants and rules.*

- *A value v and noise trades z_t.*
- *An insider who submits $x_t = X_t(v, \mathcal{H}_{t-1})$.*
- *A dealer who sets $p_t = P_t(\mathcal{H}_{t-1}, y_t)$ from the public history $\mathcal{H}_{t-1}$ and the total order flow $y_t = x_t + z_t + f_t$.*
- *Feedback cohorts whose memory states update on a public price statistic $s_t = S_t(p_0, \ldots, p_t)$ by $m_{i,t+1} = r_i m_{i,t} + s_t$ from $m_{i,0} = 0$, with $r_i = 1 - \eta_i$ and aggregate flow $f_t = \sum_i w_i m_{i,t}$, $w_i = q_i \eta_i$.*

The rules X, P and S are arbitrary causal maps. This covers Kyle, inventory-averse, learning and misspecified dealers, and any insider strategy.

(i) $f_t = \sum_{m<t} k_m\, s_{t-1-m}$, *with* $k_m = \sum_i w_i r_i^m$, *the moments of the merged measure M.*

(ii) Two cohort configurations produce the same aggregate flow for every price path on dates $t \le T$ if and only if $k_m = k'_m$ for all $m < T$. On an unbounded horizon this holds if and only if their merged measures coincide.

(iii) If $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of (v, z) the following coincide on $t \le T$: the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory. The two games therefore have the same strategy sets and the same payoff for every profile. They have the same equilibria, with the same outcomes, under any equilibrium concept defined from these payoffs and the public history.

(iv) Group-level quantities are not invariant. A Gauss and a Radau two-point configuration agree on k_0, k_1, k_2, so they are outcome-equivalent for $T = 3$, while their largest cohorts carry weights $\frac{1}{2}$ and $\frac{3}{4}$.

In words. Whatever the insider and the dealer do, two markets whose trader groups have the same merged response produce the same prices, the same order flow and the same insider and dealer outcomes. So the part of the population that prices cannot identify (Theorem 3.1) is worth nothing to anyone who trades on public information. It matters only for actions aimed at a particular group, such as halting it (Proposition 7.4). Parts (i)–(iv) are machine-checked pathwise for every causal insider, dealer and signal rule (`Paper1EquilibriumInvariance.lean`). Supplement S28 gives the proof, the unbounded-horizon statement, the extension to cohort-specific demand shocks and a numerical illustration.

Theorem 4.6 (A strategic insider in the inventory equilibrium)**.** *Add three things to the model of Theorem 4.1:*

- News. *The fundamental moves by* $F_{t+1} = F_t + \theta_t + \varepsilon_{t+1}$*, with* $\theta_t \sim N(0, \sigma_\theta^2)$ *and* ε_t *i.i.d. of variance* σ_ε^2*.*
- An insider. *At date* t *a short-lived insider privately observes* θ_t*, and nothing else, submits* d_t*, and values the position at* F_{t+1}*.*
- What cohorts and dealers see. *The cohorts trade on the observed deviation* $s_t = p_t - F_t$*. The dealers price from total flow.*

Let $\nu_t = \xi_t + d_t$ *be the unexpected flow.*

(i) Characterization. *The strategy* $d_t = b\theta_t$ *and the price* $p_t = F_t + \lambda\nu_t + x_t$*, with* x_t *the price law of Theorem 4.1, form an equilibrium exactly when the induced dynamics are stable and*

$$D = 1 - \kappa G(D), \quad \kappa(D - \bar{a}) = AVD, \quad \lambda = \frac{b\sigma_\theta^2}{\sigma_\nu^2}, \quad b = \frac{D}{2(\kappa + \lambda)},$$

$$\sigma_\nu^2 = \sigma_\xi^2 + b^2\sigma_\theta^2, \qquad V = \sigma_\varepsilon^2 + \frac{\sigma_\theta^2\sigma_\xi^2}{\sigma_\nu^2} + \left(\frac{\kappa + \lambda}{D}\right)^2 \sigma_\nu^2.$$

One unit of unexpected flow moves the price by $(\kappa + \lambda)/D$*, and the insider's expected profit per date is* $b\sigma_\theta^2/2$*.*

(ii) Kyle limit. *Without inventory risk (*$A = 0$*),* $\kappa = 0$*,* $D = 1$*,* $\lambda = \sigma_\theta/(2\sigma_\xi)$ *and* $b = \sigma_\xi/\sigma_\theta$*. These are Kyle's one-period values at every date, and the feedback cohorts do not move prices.*

(iii) Invariance. *The coefficients depend on the cohorts only through* G*, as Theorem 4.5 requires.*

(iv) Observer. *The observed deviation follows*

$$s_t = a^* s_{t-1} + \kappa^* \sum_i \tilde{u}_{i,t} + c_0 \nu_t - c_1 \nu_{t-1}, \qquad c_0 = \frac{\kappa + \lambda}{D}, \quad c_1 = \frac{\bar{a}\lambda}{D},$$

with $(a^*, \kappa^*, \tilde{u})$ *as in Theorem 4.2. This is the closed loop of Section 2 driven through a known moving-average filter* $c_0 - c_1 z^{-1}$*, which is invertible because* $c_1 < c_0$*.*

- *The memory rates remain the transmission zeros of the feedback part.*
- *The lower-bound pair of Theorem 6.1 is realized by two such equilibria with identical primitives.*
- *So the identification boundary and the* $(\Delta\eta)^{-(4K-2)}$ *rate hold with an insider present.*

In words. A short-lived insider who knows the next piece of news trades against noise traders and against the feedback traders' predictable flow. Dealers net the predictable flow out. So the insider's trades move prices through two channels: an information part λ and an inventory part, both amplified by feedback through D. Positive feedback raises the total impact. The insider then trades less and earns less, and prices reveal less of her news. Negative feedback does the opposite.

- *An example from Supplement S28.* One positive-feedback cohort lowers the share of the insider's information impounded in prices from 0.372 to 0.366. One negative-feedback cohort raises it to 0.415.
- *What prices cannot identify.* It is still worth nothing to anyone trading on public information.
- *Machine check.* `Paper1EquilibriumInsider.lean` checks the observer's pathwise reduced form, the insider's best response, the Kyle limit and the posterior variance.
- *Scope.* The insider's information is short-lived, and she conditions only on her news. Two cases are not solved: an insider who also saw the public state, who would trade against the predictable mispricing; and a long-lived Kyle insider, who would also time her trades against the cohorts' memory.

A misjudged dealer and a targeted halt. Supplement S28 adds two results in the simplest linear market covered by Theorem 4.5.

- *Rents.* A dealer that misjudges the aggregate feedback response by ε hands an insider with a committed schedule a rent of exact order ε, or charges it one (Theorem S28.1).

Public data reveal the aggregate response at the ordinary rate. So the information that protects price formation is cheap to learn, while who the groups are stays expensive.

- *A targeted halt.* Two markets that agree in equilibrium to within the cube of the memory spacing can differ at order one after their largest group is halted (Theorem S28.2). Equilibrium sharpens the case for targeted information rather than removing it.

5 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially specified input/output contract. Throughout, Φ is the standard normal distribution function.

5.1 Quantity

Theorem 5.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0,1)$. Compare a single cohort of weight q and rate η with two cohorts of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1-\epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0,\sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h^\delta_j - h^0_j)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{12}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e^*_T(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{13}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed V. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q \delta^2$. A single pulse of size

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{14}$$

and the third observation achieve error exactly α. The necessary and sufficient quantity order is δ^{-2}.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$, the necessary horizon order is δ^{-4}, attained by the pulse whenever $T \ge \max\{3, \lceil A^2/P \rceil\}$.

(v) For a fixed monetary price exposure $\lambda \neq 0$, additive reserve L and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B - L)/(|\lambda|\kappa + c)$.

In words. To tell one desk from two desks whose memories differ slightly, the observer must trade against them, and the quantity it must trade grows like $1/\delta^2$: halving the difference quadruples the trading needed. Waiting longer after a fixed trade does not help, because the trade's echo in the price dies out.

The identity in (iii) follows by direct substitution, and code checks it at $\delta \in \{0.02, 0.05, 0.10\}$: the difference is -1.2×10^{-4}, -7.5×10^{-4} and -3.0×10^{-3} at $\kappa = 1$, $q = 0.3$, and -3.2×10^{-4} at the calibration $\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 6).

A Monte Carlo check of the exact error. Formula (13) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table S32); simulated errors match the formula within Monte Carlo standard error under both hypotheses.

Figure S7 shows why the problem is hard. The two economies have the same direct impact and the same first-lag response; they first differ at lag two, by an amount proportional to δ^2, and the whole difference sequence scales with δ^2. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like δ^{-2}, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like δ^{-4}.

5.2 Execution cost and capital

Supplement S24 puts the quantity bound into money. Executing a probe against a linear book costs a positive-definite quadratic form in the probe, so no round trip yields a sure gain (Lemma S24.1). A counterparty that fills the probe needs margin that grows with the square of the probe, so the capital needed grows as $1/\delta^4$: halving the difference between two memories multiplies it by sixteen (Theorem S24.2); the same order holds for capital paid by the probe trader (Lemma S24.4). If the observer can instead see an account's exact equity, one test trade settles the question (Proposition S24.3): the difficulty lies in what the observer is allowed to see, not in the market.

A worked example with two Kalman desks that differ only in their signal-to-noise ratios (Supplement S15) puts these orders into numbers.

6 Unknown mixtures and passive data

Theorem 5.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering them grows.

Two facts from the case of two cohorts carry over (Supplement S25). Fitting poles to the price response targets the closed-loop eigenvalues and misses the memory rates, which are zeros of the transfer function. And for two unknown groups no estimator, biased or not, separates them with fewer than order $(\Delta\eta)^{-6}$ experiments (Theorem S25.1), while the two hardest-to-separate economies have the same summed price response, so what prices cannot see carries no market-wide risk (Remark S25.2). The general case follows.

6.1 Many unknown cohorts

Theorem 6.1 (K-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, so that all rates lie in $(0,1)$, and assume $a + \kappa W/(1 - r_c - \Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes $T \geq 2K$ noisy taps ($T = \infty$ allowed); the bound below uses the full response norm, which dominates any finite prefix. There exist two K-cohort economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \ldots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from N independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem 6.2 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which Theorem S10.1 matches.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight

function: both are positive, have K atoms and integrate polynomials of degree $2K-2$ exactly. For $K = 2$ it reproduces the order of Theorem S25.1. Table S33 lists the two rules: their low moments agree to machine precision and their moment of order $2K-1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent. The two rules also differ in their largest weight for every K we checked: for $K = 2, \ldots, 200$ the largest Gauss–Radau weight exceeds the largest Gauss weight, by 0.25 at $K = 2$ and by about $0.79/K^2$ for large K (`equilibrium/gauss_radau_maxweight.py`). So the argument given for $K = 2$ in Proposition 7.4 applies to each such K. A proof for every K is open.

What counts as an experiment. In Theorem 6.1 an experiment is a unit impulse from the reset state whose response is observed with noise of variance σ^2 per tap. The exponent is the same for any probe design of bounded energy. For any probe v the mean difference between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy E each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and Supplement S9 prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section 7).

Table S34 and Figure S8 confirm the order numerically.

Theorem 6.2 (Matching upper bound for every K)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\neq j} |s_i - s_j| \geq \gamma > 0$, and merged weights $w_i \in [\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments whose responses carry Gaussian noise with variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(4K/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(4K/\beta)}\;\epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence. On any such class that contains the Gauss and Gauss–Radau pair of Theorem 6.1 (a rich class), together with that theorem the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$. On smaller classes the upper bound need not be tight: if the rates are known, for example, the weights need only order $\sigma^2(\Delta\eta)^{-2}$ experiments. The constants c_0

and C grow quickly with K, through the conditioning of the moment map; the theorem gives the order in $\Delta\eta$ for fixed K, not a usable constant for large K (in Supplement Table S4 our double-precision Prony implementation already fails at $K = 4$ on the disclosure grid).

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ (plus κ at $t = 0$) can be solved for $k_0, \ldots, k_{2K-1}$ from $h_1, \ldots, h_{2K}$. Because $h_0 = \kappa$ is a known primitive, each k_m is a polynomial in $h_1, \ldots, h_{m+1}$ with coefficients depending on (a, κ) (the only division is by $\kappa h_0 = \kappa^2$), so on the bounded set of responses within sup-distance 1 of the class this map is L_h-Lipschitz, with L_h depending on the class but not on $\Delta\eta$. Let E be the event $\max_{1 \le t \le 2K} |\bar{h}_t - h_t| \le \sigma N^{-1/2} \sqrt{2 \log(4K/\beta)}$, where $\bar{h}_t$ averages the N responses; a Gaussian tail bound and a union bound over the $2K$ estimated taps give $P(E) \ge 1 - \beta$. On E, and provided this radius is at most 1, the estimated kernel values satisfy $\max_m |\hat{k}_m - k_m| \le L_h \sigma N^{-1/2} \sqrt{2 \log(4K/\beta)}$. Centring at r_c is a fixed triangular linear map, $\mu_m = \sum_{j \le m} \binom{m}{j} (-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \ldots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j} (s_i - s_j)^4$, bounded away from zero on the compact class. On ordered, γ-separated nodes F is injective, because a measure with K distinct atoms is determined by its first $2K$ moments. A C^1 map that is injective on a compact set Q and has a nonsingular Jacobian there is injective on an open neighbourhood U of Q: otherwise there are $x_n \ne y_n$ with $F(x_n) = F(y_n)$ and distance to Q tending to zero, whose limits $x, y \in Q$ satisfy $F(x) = F(y)$, so $x = y$, contradicting local injectivity at x. Applying the quantitative inverse function theorem uniformly on Q, there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F(U)$ contains the ball of radius ρ_0 around every point of $F(Q)$ and the single inverse $F^{-1} : F(U) \to U$ is L-Lipschitz on each such ball. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K - 1$. Since $\Delta\eta < 1$ and the centring map is fixed, on E we get $\|\hat{t} - t\|_\infty \le C_1 \sqrt{\log(4K/\beta)}\, \epsilon_N$ with C_1 equal to $\sqrt{2} L_h$ times the norm of the centring map; and because $\sigma N^{-1/2} \sqrt{2 \log(4K/\beta)} \le \sqrt{2}\, \epsilon_N \sqrt{\log(4K/\beta)}$, the radius condition holds whenever $c_0 \le 1/\sqrt{2}$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. For an input in such a ball, $x = F^{-1}(\hat{t})$ has K distinct nodes and nonzero weights, so the $K \times K$ Hankel matrix of $\hat{t}$ is nonsingular and Prony's method returns exactly x: the Prony polynomial is determined uniquely by the Hankel system, and since the K real nodes of x reproduce $\hat{t}$ it equals $\prod_i (z - \hat{s}_i)$, so its roots are real and distinct and no complex pair can appear; Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C \sqrt{\log(4K/\beta)}\, \epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = L C_1$ and $c_0 = \min\{\min(\rho_0, \gamma/(2L))/C_1,\, 1/\sqrt{2}\}$, which also keeps $C \sqrt{\log(4K/\beta)}\, \epsilon_N < \gamma/2$. No

separate root-perturbation bound is needed: on the ball where F^{-1} is L-Lipschitz, Prony's roots and weights are the coordinates of $F^{-1}(\hat{t})$, so the conditioning of the root step is part of L (for an explicit step-by-step analysis of Prony's method with clustered exponents see Katz et al., 2023). The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 6.1 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

In words. With K unknown groups whose memories lie within $\Delta\eta$ of each other, the number of experiments needed grows like $(\Delta\eta)^{-(4K-2)}$. For two groups, halving the spread multiplies the data needed by 64; for three groups, by 1,024. Theorem 6.1 shows that no method does better, and this theorem shows that Prony's method achieves it.

Machine check. Lean verifies this theorem for every $K \geq 1$, using only Lean's standard axioms. The formal proof follows the one above: a Lipschitz bound for the closed-loop deconvolution that does not depend on $\Delta\eta$, the $(\Delta\eta)^{-m}$ rescaling, a sub-Gaussian tail bound over the $2K$ taps, injectivity of the Jacobian of the moment map (proved by Hermite interpolation, without the determinant formula), and a uniform inverse function theorem on the compact class. The formal estimator is the representation with increasing nodes and nonzero weights that matches the $2K$ estimated moments; Lean shows that it is unique and that Prony's linear system has a unique solution whose polynomial has exactly its nodes as roots, so the roots are real. The formal statement allows sub-Gaussian noise and assumes $|r_c| + \Delta\eta \leq 1$, so that every memory rate has modulus at most one.

Corollary 6.3 (Unknown primitives cost nothing in the rate)**.** *In the setting of Theorem 6.2, suppose in addition that every memory rate is at least $\varepsilon > 0$ (that is, $r_c - \Delta\eta \geq \varepsilon$), and that the fundamental value p^*, the impact $\kappa \in [\underline{\kappa}, \overline{\kappa}] \subset (0, \infty)$ and the persistence $a \in [0, \bar{a}] \subset [0, 1)$ are unknown. Put $\tilde{k}_0 = k_0 + a/\kappa$ and $\tilde{k}_m = k_m$ for $m \geq 1$.*

(i) The response depends on (a, κ, M) only through κ and $\tilde{k}$: $h_0 = \kappa$ and $h_{t+1} = \kappa \sum_{m=0}^{t} \tilde{k}_m h_{t-m}$. The persistence a acts exactly like an extra group with memory rate zero and merged weight a/κ.

(ii) The taps $h_0, \ldots, h_{2K+1}$ identify (κ, a, M): $\kappa = h_0$; the recursion gives $\tilde{k}_0, \ldots, \tilde{k}_{2K}$; the values $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of $\mu' = \sum_i w_i r_i \delta_{r_i}$; and $a = \kappa(\tilde{k}_0 - \sum_i w_i)$.

(iii) Subtract the pre-probe price to remove p^* (doubling the noise variance), estimate $\hat{\kappa} = \hat{h}_0$, deconvolve with $\hat{\kappa}$ alone, apply Prony's method to the estimated moments of μ', and

set $\hat{w}_i = \widehat{w_i r_i}/\hat{r}_i$ *and* $\hat{a} = \hat{\kappa}(\hat{\tilde{k}}_0 - \sum_i \hat{w}_i)$. *There are constants* $c_0, C > 0$ *depending on* $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, \varepsilon, \underline{\kappa}, \overline{\kappa}, \bar{a}, r_c)$ *but not on* $\Delta\eta$ *such that, whenever* $\epsilon_N \le c_0$,

$$\max_i |\hat{w}_i - w_i| + |\hat{a} - a| + |\hat{\kappa} - \kappa| \;\le\; C\sqrt{\log(8K/\beta)}\,\epsilon_N$$

with probability at least $1 - \beta$. *Hence* $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ *experiments suffice, the same order as with known primitives.*

In words. Not knowing the market's own persistence or its price impact costs a constant factor, not a worse rate. Persistence enters the price exactly like one more group of traders with no memory at all, and that group sits far from the clustered groups the supervisor is trying to separate. Measured in the merged weights times the rates, the clustered groups' problem is the same Prony problem as before, with weights bounded below because rates are bounded away from zero; one more tap pins down the extra group.

Proof of Corollary 6.3. (i) Substitute $k_0 = \tilde{k}_0 - a/\kappa$ in $h_{t+1} = ah_t + \kappa \sum_{m \le t} k_m h_{t-m}$: the term ah_t cancels against $\kappa(a/\kappa)h_t$. (ii) Solving the recursion of (i) for $\tilde{k}_t$ gives $\tilde{k}_t = (h_{t+1} - \kappa \sum_{m<t} \tilde{k}_m h_{t-m})/\kappa^2$, a polynomial in $h_1, \dots, h_{t+1}$ divided by a power of $h_0 = \kappa$. For $m \ge 1$, $\tilde{k}_m = \sum_i w_i r_i^m = \sum_i (w_i r_i) r_i^{m-1}$, so $\tilde{k}_1, \dots, \tilde{k}_{2K}$ are the moments of orders $0, \dots, 2K-1$ of μ', which has K distinct atoms with positive weights; these moments determine μ' (Theorem 3.1). Since $r_i \ge \varepsilon > 0$, $w_i = (w_i r_i)/r_i$, and then $a = \kappa(\tilde{k}_0 - \sum_i w_i)$. (iii) On the event that every estimated tap is within $\sigma N^{-1/2}\sqrt{2\log(8K/\beta)}$ of its mean (probability at least $1 - \beta/2$ by the Gaussian tail and a union bound over $2K+2$ taps, with the doubled variance absorbed into the constant), $\hat{\kappa} \ge \underline{\kappa}/2$ once $\epsilon_N \le c_0$, so the map from taps to $\hat{\tilde{k}}$ is Lipschitz with a constant that depends on $(K, \underline{\kappa}, \overline{\kappa}, \bar{a}, W)$ and not on $\Delta\eta$. The weights of μ' lie in $[\varepsilon\omega_{\min}, \omega_{\max}]$ and its rates are those of the class of Theorem 6.2; the proof of that theorem uses only a $\Delta\eta$-free Lipschitz deconvolution, the rescaling by $(\Delta\eta)^{-m}$ and weights bounded below, so it applies verbatim to μ' and gives $\max_i |\widehat{w_i r_i} - w_i r_i| \le C_1\sqrt{\log(8K/\beta)}\,\epsilon_N$ and rate errors smaller by a factor $\Delta\eta$, with probability at least $1 - \beta/2$. The maps $(v, r) \mapsto v/r$ on $r \ge \varepsilon/2$ and $(\kappa, \tilde{k}_0, w) \mapsto \kappa(\tilde{k}_0 - \sum_i w_i)$ are Lipschitz on the bounded set reached, and $|\hat{\kappa} - \kappa|$ and $|\hat{\tilde{k}}_0 - \tilde{k}_0|$ are of order $\sigma N^{-1/2} \le \epsilon_N$. Combining gives the bound. □

Table S35 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), and Katz et al. (2023) show that Prony's method attains the min-max bound for clustered exponents; here the nodes are real and are first

extracted from the closed loop, and it reproduces the ESPRIT slope of -6.15 reported for $K = 2$ in the earlier version.

Real market noise. Theorems 6.1 and 6.2 assume Gaussian noise. A pre-specified test (Supplement S16) keeps the trader groups synthetic and replaces the noise with heavy-tailed one-minute Bitcoin price changes: the price-only requirement still grows with fitted slopes -6.01 for $K = 2$ and -9.81 for $K = 3$ (theory -6 and -10), while the tagged requirement stays bounded across the spacing grid.

General memory. If memory is any completely monotone kernel, a finite or infinite mixture of exponential memories, recovery becomes far harder: the minimax accuracy for the memory distribution is of exact order $1/\log(N/\sigma^2)$, so reaching accuracy w requires and suffices $N/\sigma^2 = \exp(\Theta(1/w))$ (Theorem S17.2, Supplement S17). The polynomial rates above rely on memory being a finite mixture of exponentials.

6.2 Passive data

Theorem 6.4 (Passive identification: minimum phase)**.** *Consider a stable economy with $0 < a < 1$, distinct memory rates $0 < r_i < 1$ and nonzero merged weights.*

(i) The zeros of H are the memory rates r_i together with a structural zero at $z = 0$; none cancels against the denominator, and all lie in the open unit disk, so H is minimum phase. The degenerate anchor ($a = 0$) and zero-rate cases require the cancellations described in the proof of Theorem 3.1.

(ii) Let the price be driven by a single innovation sequence with variance $\sigma_\xi^2 > 0$, possibly unknown, so that its spectrum is $f(\omega) = \sigma_\xi^2 |H(e^{i\omega})|^2/(2\pi)$. Within the class of (i), f determines $\sigma_\xi^2\kappa^2$ and the normalized response H/h_0 (recall $h_0 = \kappa$); hence it determines the memory rates. If in addition κ (or σ_ξ^2) and a are known, it determines H and, by Theorem 3.1, the merged measure.

Machine check. For every K, Lean checks part (i) and part (ii) starting from equal spectra at every frequency (`p1_thm_minphase_zeros`, `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged`). The check uses H in the rational form $\kappa z D(z)/((z - a)D(z) - \kappa N(z))$ of Theorem 3.1, and *stable* in the sense that the poles of H lie in the open unit disk. Both are derived from the market recursion. The eigenvalues of the transition of the state $(x_t, y_{1,t}, \ldots, y_{K,t})$ are exactly the zeros of the denominator of H, so Schur stability of the recursion is the pole condition (`p1_eigen_iff_bottom_root`,

`p1_schur_stable_iff_poles_in_disk`). The generating function of the recursion's price response h_t to a unit trade satisfies $\left(\sum_t h_t x^t\right) x^{K+1} B(1/x) = x^{K+1} \kappa x^{-1} D(1/x)$ as formal power series, where B is the denominator of H; with $x = 1/z$ this is $\sum_t h_t z^{-t} = H(z)$ (`p1_impulse_generating`, `p1_tfBottomRev_eval`, `p1_tfTopRev_eval`).

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 6.5 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem S25.1, $D = \Theta(\tau^2 (\Delta\eta)^6)$.

(iii) For every $K \geq 2$, for the K-cohort Gauss and Gauss–Radau economies of Theorem 6.1, $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of cohorts. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate D (Stein's lemma), and both errors decay at the Chernoff-information rate, which is of the same order in the separation; so the orders are the right ones; we do not claim that this attainment is uniform as the separation shrinks. All constants depend only on K, the stability margin of the family, the rate interval $[\eta_{\min}, \eta_{\max}]$, the total weight and the moment-matched construction; they do not depend on δ, $\Delta\eta$ or T. In part (ii) τ is held fixed as $\Delta\eta \to 0$. The testing conclusions use a finite-record bound $D_{\mathrm{KL}}(P_1^T \| P_0^T) \leq C\, T\, \varepsilon^2$, with ε the leading moment gap (δ^2, $\tau(\Delta\eta)^3$, $(\Delta\eta)^{2K-1}$); the lower bounds on D and the fixed-pair error exponents are separate statements.

Machine check. Lean checks the finite-sample consequence of parts (i) and (iii), not the divergence-rate statements. In the checked record the market starts from rest

and runs for any number M of warm-up periods before the T recorded prices. The bound does not depend on M, but the passage to the stationary law ($M \to \infty$) is not checked. Part (iii) is checked for every pair of populations with nonnegative merged weights of total at most W, normalized rates in $[-1, 1]$ and equal moments below order $2K - 1$ (`p1_thm_passive_cost_matched`). This covers the K-cohort Gauss and Gauss–Radau economies of Theorem 6.1: both rules are positive with nodes in $[-1, 1]$ and integrate polynomials of degree $2K - 2$ exactly. Lean proves these properties of the two rules for $K = 2$ and $K = 3$ (`p1_thm_passive_cost_gauss_radau_two`, `p1_thm_passive_cost_gauss_radau_three`), so for these K the check covers the paper's economies with no extra hypothesis. For general K, the exactness of the two quadrature rules, a classical property, enters the check as the moment hypothesis. It also covers the binomial pair used in the numerical checks (`p1_thm_passive_cost_every_K`).

In words. Watching the market without trading does not escape the barrier. A passive observer needs at least the same polynomial order of observations as a prober needs experiments; it pays in time rather than in trading losses. Prices recorded with extra measurement noise that is independent of the price are a garbling of the noiseless record, so by the data-processing inequality they carry less information, and the lower bound holds a fortiori.

The proof (Supplement S1) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = z^{-1} H_1 H_0 (G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order δ^2 for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are -4.21 and -5.93, against the theoretical -4 and -6; for part (iii) the fitted slopes of the Whittle divergence rate in $\Delta\eta$ are 5.91, 9.98 and 14.05 for $K = 2, 3, 4$, against 6, 10 and 14.

Passive surveillance is possible in principle but needs at least the same polynomial order of observations as active probing needs experiments. Proposition S3.2 is the exact finite-sample version for the known pair.

7 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- I_0: public prices only.
- I_1: prices and aggregated executed flow.

- I_2: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- I_3: prices and exact audited equity (Proposition S24.3), also with reports rounded to 0.01.
- I_4: algorithm update timestamps without the update values.
- I_5: a supervisory audit registry with access to internal algorithm states.

Table 2 shows that disclosure is not one thing. Timestamps without update values carry no information about δ in this model. Noisy periodic statements barely help. Exact audited equity collapses the requirement to a single trade, limited only by lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

Proposition 7.1 (Tagged flow removes the barrier)**.** *Let the price x_t be the causal stationary solution of the model driven by i.i.d. centred innovations of variance $\sigma_\xi^2 \in (0,\infty)$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose cohort i has merged weight $w_i > 0$ and the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. centred measurement noise of finite variance σ_f^2, independent of the whole innovation sequence and hence of the whole price path. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top) \big], \quad (15)$$

where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s+m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \quad (16)$$

V_i depends only on cohort i's own (w_i, r_i), σ_f^2 and the price autocovariances. Two conclusions are distinct: (a) for each admissible economy, $\hat{\beta}_i$ has the stated limiting normal law; (b) on a uniformly stable family with stationary price variance at least $\gamma_{\min} > 0$ and merged weights at least $w_{\min} > 0$, the limiting covariance V_i is bounded by a constant that does not depend on $|r_i - r_j|$. Statement (b) bounds the limit; it is not a central limit theorem that holds uniformly over the family. Three pieces are machine-checked: the finite-sample reset-experiment version, with an error bound that does not depend on the spacing; the lower

bound on the determinant of the population instrument matrix that does not depend on the spacing (`p1_tagged_iv_determinant_bound`); and the exact decomposition of the IV score into reporting-error terms with coefficients fixed by the price path plus two boundary terms (`p1_iv_boundary_decomposition`). The stationary law of large numbers and the central limit theorem behind (a) are not machine-checked.

In words. If each group's trading is reported separately, each group becomes an ordinary regression. Its precision depends on the group's own rate and on the price dynamics, but it has no singularity as the groups' rates approach each other.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates; the closed form (16) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with 4×10^5 periods). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

7.1 How much disclosure is worth

Proposition 7.1 says that tagging removes the barrier; the next result says by how much.

Theorem 7.2 (Value of class-tagged disclosure)**.** *Take a rich class of Theorem 6.2: rates $r_c + \Delta\eta\, s$ with s in a compact set of γ-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem 6.1, and merged weights in a nondegenerate interval $[\omega_{\min}, \omega_{\max}]$, $0 < \omega_{\min} < \omega_{\max}$. In each reset experiment the observer applies a unit impulse and records the price response with independent $N(0, \sigma^2)$ noise per tap; under tagging it also records each class's flow response $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$ with independent $N(0, \sigma_f^2)$ noise per tap. Let N_p and N_t be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants $0 < c \le C < \infty$, depending on (σ, σ_f) and the class but not on $\Delta\eta$, such that*

(i) to recover the weights to a fixed accuracy smaller than half the weight difference between the Gauss and Gauss–Radau economies (so that recovering the weights decides between them), $c \le N_t \le C$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-2)}$;

(ii) to recover the rates to accuracy $c_0\Delta\eta$, for a fixed $c_0 > 0$ smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target), $c(\Delta\eta)^{-2} \le N_t \le C(\Delta\eta)^{-2}$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-4)}$.

Proof. $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 6.1 and 6.2. *Upper bounds.* Average the N price responses to estimate h with error of order $\sigma N^{-1/2}$ per tap. Given h, class i's mean flow is a two-parameter family $F(w_i, r_i; h)$ whose Jacobian columns $\sum_s r_i^{t-s} h_s$ and $w_i \sum_s (t-s) r_i^{t-s-1} h_s$ are linearly independent, with a Gram determinant bounded below uniformly on the class because $w_i \geq \omega_{\min}$, the rates lie in a compact subset of $(0, 1)$ and $h_0 = \kappa > 0$; the bound does not involve the other classes or $\Delta\eta$. Least squares on class i's flows with the plug-in h therefore estimates (w_i, r_i) with error $O_P(N^{-1/2})$, uniformly on the class, which gives the upper bounds in (i) and in (ii) for accuracy $\epsilon\Delta\eta$. For (i) there is also a finite-sample version: in the first period the tagged flow of class i is $w_i\kappa$, so the mean of the N first-period records gives $P(|\hat{w}_i - w_i| \geq \epsilon) \leq \sigma_f^2/(N\kappa^2\epsilon^2)$ by Chebyshev's inequality, for every configuration (this step is machine-checked). *Lower bounds.* At a configuration in the interior of the class, perturb one class's weight by ϵ, or its rate by $\epsilon\Delta\eta$; for small ϵ both perturbed configurations remain in the class because the weight interval is nondegenerate and the rate set has nonempty interior. The mean differences of both the price and the flow records are of order ϵ, respectively $\epsilon\Delta\eta$, so the Kullback–Leibler divergence of N experiments is at most $CN\epsilon^2$, respectively $CN\epsilon^2(\Delta\eta)^2$, and Le Cam's two-point lemma gives the lower bounds. □

In words. This is the value of disclosure measured in data. Tagging replaces a requirement that explodes as the groups become similar with one that does not grow at all (for the weights) or grows only like $(\Delta\eta)^{-2}$ (for the rates).

Proposition 7.1 and Theorem 7.2 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

Proposition 7.3 (Conditional limits of three evasion arguments)**.** *Maintain the exogenous reporting-noise and valid-instrument assumptions of Proposition 7.1; fix the finite number of linked class labels and a uniformly stable model family.*

(i) For a fixed split $u_{i,m} = \alpha_{i,m} u_i$ with known common ownership, independent mean-zero reporting errors and $\alpha_{i,m} \geq \underline{\alpha} > 0$, only the lagged-flow column of the population IV matrix scales. Thus $\Sigma_{i,m} = \Sigma_i \operatorname{diag}(\alpha_{i,m}, 1)$ and $\det \Sigma_{i,m} = \alpha_{i,m} \det \Sigma_i$. Subject to the remaining uniform moment and excitation conditions, fixed finitely many consistent sub-account estimates can be summed: the true identity is $w_i = \sum_m w_{i,m}$, and the estimated sum has sampling error. The share restriction implies $M_i\underline{\alpha} \leq 1$. This is not robustness to arbitrarily many vanishing accounts; a common estimated rate does not identify ownership.

(ii) Assume separately an execution cost lower bound $\mathbb{E}C_\zeta \geq c_\zeta \sigma_\zeta^2$ *with* $c_\zeta > 0$ *and a gross profit upper bound* $\mathbb{E}\Pi \leq \overline{\Pi}$ *uniform in the spacing. Camouflage is unprofitable when* $c_\zeta \sigma_\zeta^2 > \overline{\Pi}$. *A variance requirement of order* $(\Delta\eta)^{-(4K-2)}$ *requires an additional additive exogenous-noise estimation model with uniformly controlled instruments. Neither this variance requirement nor the execution bound is established for arbitrary strategic orders.*

(iii) If intensity is scaled to $\gamma_i w_i$ *while* $1 - r_i \geq \eta_{\min} > 0$, *then* $q_i \leq \gamma_i w_i / \eta_{\min} \to 0$ *as* $\gamma_i \to 0$. *The other groups remain in the aggregate budget; reducing one contribution does not itself establish that the ceiling is met.*

Economic scope. These are conditional calculations about fixed linked accounts, bounded execution costs and a persistence class. They do not solve an equilibrium game of evasion. In particular, endogenous orders can affect prices and invalidate the reporting-error instrument, and a persistence change can offset a reduction in weight. Adversarial account creation and response adaptation require further modelling.

In a model calculation using the same assumed noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of -6.08 for prices and -4.34 for the ratio at $K = 2$ (theory -6 and -4). Calibrated to the descriptive persistence estimates below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972). These rates come from day-demeaned regressions, which bias persistence down (Supplement S19), so the factors are illustrative.

7.2 When disclosure is worth acquiring

When is disclosure worth its cost? In a stylised cost comparison (Proposition S27.1, Supplement S27), a supervisor prefers an audit trail when the harm from slow attribution exceeds the trail's cost per market. Disclosure is most valuable in a middle band of memory spacings, where prices are too slow and labels are fast, and only when the trail covers enough markets to share its fixed cost.

A decision in which attribution matters. A supervisor chooses between a uniform reduction in activity and halting the largest verified class, over a declared supervisory horizon. The target below is the feedback budget $S = a + \kappa \sum_i q_i$, not the spectral radius of the state

matrix. In an already stable positive economy, imposing $S \le \bar{s} < 1$ is a tighter prudential constraint, not restoring stability to an unstable market.

Proposition 7.4 (Supervisory separation: market-wide versus targeted intervention)**.** *Assume the positive stable model, known a, κ, and $a < \bar{s} < 1$. Write $q_i = w_i/(1 - r_i)$ and $R = S - \kappa \max_i q_i$ for the post-halt feedback budget.*

(i) A uniform scale θ meets the ceiling exactly when $a + \theta(S - a) \le \bar{s}$. Its largest feasible value is $\theta^* = \min\{1, (\bar{s} - a)/(S - a)\}$. The identity $S = 1 - \kappa/\sum_j h_j$ is algebraic; estimating this infinite sum requires a justified tail bound and an observation/noise model. It alone establishes no unconditional parametric learning rate.

(ii) For $K = 2$, the Gauss and Gauss–Radau pair of Theorem 6.1 has limiting largest normalized weights $1/2$ and $3/4$. Thus $|S^G - S^R| = O((\Delta\eta)^3)$ and $q^R_{\max} - q^G_{\max} = W/(4(1 - r_c)) + O(\Delta\eta)$. For sufficiently small spacing, a nonempty interval of ceilings separates their post-halt decisions. For N independent reset records with common Gaussian noise $\sigma^2 I$, both decision errors at most $1/4$ require

$$N \ge (2\Phi^{-1}(0.75)\sigma/\|h^G - h^R\|_2)^2 \ge c\sigma^2(\Delta\eta)^{-6}.$$

The record horizon and the response norm must be the same on both sides.

(iii) Tagged-flow estimation under Theorem 7.2, a rate bound away from one, and a fixed nonzero decision margin gives a bounded sample requirement as the spacing shrinks. The target class must be mapped to an executable, verified class label; a largest latent atom alone supplies no such identity.

The proof and the general-K scope are in Supplement S29.

Robustness, costs and the bridge to data. Supplement S29 adds a condition under which strategic responses cannot overturn the separation, a complete cost-and-error model with a robust disclosure gate (Proposition S29.1), an operational bridge from tagged flows to a halt decision, and the condition under which buying information beats the best no-data action (Proposition S29.2).

What the data can and cannot test. The theorems concern experiments that no outside researcher can run: reset initial states, known price persistence and impact, and either controlled probes or stationary data from a known innovation law. Public data meet none of these exactly. Order flow is endogenous, the fundamental value is unobserved and replaced by returns, observations overlap in time, parameters drift, and the class reports are positions

aggregated over many accounts rather than tagged flow. The tests below therefore check the theory's observable prediction, that class records restrict persistence which prices leave unrestricted, under a stated mapping from class positions to cohort states. They do not estimate the theorems' rates, and a pass does not show that class persistence is learning memory.

7.3 Evidence from trader-class reports

Two public data sets report positions by trader class: the CFTC's weekly Commitments of Traders, mandated by law, in twenty regulated futures markets, and Binance's five-minute top-trader and all-account ratios, published voluntarily, in 30 perpetual markets. Supplements S18 and S19 report the pre-specified tests in full; the formal two-root stock–flow inventory bridge linking Remark 2.2 to reported position stocks and flows is derived at the beginning of Supplement S18. In both datasets, class positions pin each class's fitted persistence to a narrow interval while prices leave it essentially unrestricted: the primary CFTC hypothesis holds in all eight markets, with price-only intervals 9 to 71 times wider, and in all twelve markets of a later registered replication; matching the weekly report's precision from prices alone would take, at the median, about 1,350 times as much data under standard root-T scaling (a hypothetical conversion, not an identified data requirement; Table S26). On 30 fresh Binance markets the horizon-matched comparison gives a median width ratio of 7.3. Several pre-specified tests failed and are reported as they came out (Table S29): Binance class data did not improve out-of-sample return forecasts, the CFTC reports read as flows pinned down their own persistence in only 3 of 8 markets (6 were required), and a test designed to separate learning memory from position turnover found a distinct faster root in only 2 of 8. These failures are consistent with several explanations, including within-class sign cancellation: coarse reporting buckets (e.g., CFTC Leveraged Funds) mix offsetting strategies (such as cash–futures basis arbitrageurs trading against trend-following momentum CTAs), motivating tests of granular algorithm-tagged records without proving their necessity.

8 Implications for supervision and market design

The results separate two supervisory tasks that are often run together. Monitoring aggregate fragility, for example to set market-wide limits, needs only the combined price response, which public data can identify under the observation and tail conditions of the model (Remark S25.2, Proposition 7.4(i)). Attributing feedback to particular groups, to apply targeted

limits or enforcement, faces the $(\Delta\eta)^{-(4K-2)}$ barrier whenever groups react over similar horizons (Proposition 7.4(ii)). Class-tagged records remove that barrier group by group (Proposition 7.1), and in the model persistent pseudonymous class labels deliver the full statistical gain without personal data.

What a supervisor can learn from public data. Prices and anonymous flow reveal the merged measure M and the closed-loop impulse response, hence the transfer function, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close. This is not a failure of econometric technique: Theorem S25.1 holds for every estimator, biased or not.

What probing costs. At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about 10^5 USDT of capital. Probing is therefore feasible for coarse questions and infeasible for fine ones. Because capital grows as δ^{-4} and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by 2^{4K-2} for K unknown cohorts.

What disclosure buys. Under the model's stated observation contracts, verified class-tagged executed flow can remove the spacing penalty for fixed-accuracy weight recovery (Proposition 7.1). The attribution experiment compares those records with specified periodic statements; its ranking does not apply to every possible disclosure design. The factor of about 333 obtained from Binance's day-demeaned ratio persistences is an illustrative model calibration. Top traders and all accounts overlap, and the published ratios are not the disjoint executed-flow records used by the theorem. CFTC position reports likewise provide descriptive information about fitted persistence, without validating the theorem's observation contract or identifying algorithmic memory.

When audit trails already exist. The US Consolidated Audit Trail (SEC Rule 613) is designed to capture specified order and reportable-event information for NMS stocks and listed options across exchanges and broker-dealers, from origination to execution or cancellation, linked to a customer identifier and to the broker-dealer that handled the order; its actual scope is governed by Rule 613, the CAT NMS Plan and exemptions. Since February 2025, under SEC exemptive relief, it no longer requires the names, addresses and years of birth of natural-person customers, who remain linked through an identifier derived from a transformed tax identifier (SEC, 2012, 2016, 2024, 2025). Our results bear on such regimes in

three conditional ways. First, for the attribution task modelled here, records that report each class's flow separately remove a barrier that, for an observer who learns by probing, costs round-trip trading loss of order δ^{-4}, and they save samples by a factor of order $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy or $(\Delta\eta)^{-(4K-4)}$ for rates resolved to a fraction of their spacing (Proposition 7.1, Theorem 7.2). This is a measurement value, not a cost–benefit assessment: CAT's budget pays for enforcement, market reconstruction and surveillance, none of which the model measures, and Supplement S8 sets the hypothetical probing loss beside it only as a scale illustration. For estimating class responses in our model a persistent pseudonymous label suffices; this does not settle what enforcement or data-minimization design requires.

Second, an audit trail may connect accounts, entities and orders without identifying a homogeneous decision rule. Within-class heterogeneity and opposite-signed responses can produce cancellation, but the reported regression failures do not identify cancellation as their cause. Finer strategy or algorithm labels could help only if membership, timing, executed quantities and the reporting-error model meet the identification assumptions. An algorithm identifier does not by itself verify a fixed linear response, ownership linkage or persistence. The paper therefore establishes neither that a particular reporting granularity is necessary for every supervisory task nor that finer identifiers automatically restore a parametric learning guarantee.

Third, the audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not public: regulators and authorized self-regulatory organizations have controlled access, while outside desks and researchers generally do not. For these observers the price-only and coarse-disclosure contracts of this paper are the relevant ones.

What to monitor instead. Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem S12.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and confidence sets for the transfer function, both of which are identified from public data. These are cheap where identification is expensive (Remark S25.2); the companion paper shows that the summed response can be bounded at the parametric rate from controlled experiments, and only under a declared bound on memory length from a finite window (Niroula et al., 2026).

How to measure feedback. Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and

(iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

9 Conclusion and an open problem

In a linear market with exponential-memory cohorts and known impact parameters, prices reveal a merged summary of algorithmic feedback. Resolving the model cohorts costs executed quantity of order δ^{-2} and trading capital of order δ^{-4} for a known pair, with a numerically evaluated constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for K unknown cohorts, and at least that order of passive observations; in simulations of stylized nonlinear rules and of online learners trained in the closed loop the same costs hold when the competing populations' feedback weights match, populations whose learned weights differ are told apart by those weights instead, and hard thresholds hide cohorts from small probes altogether. Without the exponential form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-tagged disclosure removes the barrier under the stated contract, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$ for resolving rates (and $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy), an illustrative model-calibration factor of about 333 using the published BTCUSDT ratio persistences, without establishing that the actual reporting series meet the theorem's observation contract. The CFTC's class reports restrict a fitted position persistence that prices leave unrestricted in all eight regulated markets we examined; this persistence is not shown to be learning memory, a pre-specified test of what prices do reveal failed, and class-tagged memory states did not improve 30-minute post-shock return forecasts ($\Delta R^2 = -1.09$ points, inconclusive). Differencing weekly CFTC position stocks under an additional stock–flow interpretation gives descriptive response regressions, which failed a pre-specified precision test, and a pre-specified test for a second, faster root in positions, which would have been the learning memory, failed as well. With real Bitcoin price noise in place of Gaussian noise, the predicted rates held in a pre-specified test, at the sample sizes where the real noise could be used directly. Under the specified Roll-noise calibration the model capital bound for a moderate separation is about 10^5 USDT; this is not a measured real-market identification bill. Estimated feedback associations in crypto markets are small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance. In an equilibrium with inventory-averse dealers, price impact is the price of inventory risk and persistence is the dealers' layoff speed. Without feedback this equilibrium is the paper's closed loop. The equilibrium price is the

same closed loop with effective parameters and unchanged memory rates, so the identification boundary and the rates hold in equilibrium. Positive feedback raises impact, and an equilibrium exists only below a risk-aversion threshold that falls in inverse proportion to the feedback intensity. Economies with the same merged measure are outcome-equivalent for every causal insider and dealer, so what prices cannot identify matters only for actions aimed at a particular group.

Two open questions remain. The first is economic: Proposition 7.4 gives one stylised decision in which knowing the individual groups changes the right action, but turning the information costs here into a value of disclosure needs a supervisory loss function estimated from real interventions. The second is for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response and the transfer function, and finite-sample confidence sets for them, are identified from public data even when cohorts are not. The structural matrix J is not: economies with the same merged measure can have different state dimensions, so J is identified only up to the equivalence class of minimal realizations of the same transfer function. A well-posed next question is: for a stated class of stable closed-loop markets, given an estimated transfer function and a finite-sample confidence set, which intervention policy of least cost guarantees a stated bound on future price displacement without recovering individual cohorts? The companion paper uses exactly this interface to build an identification-free pre-trade control.

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Data availability. All data are public. Binance market data come from the Binance public data archive (`https://data.binance.vision`): spot and USD-margined perpetual futures candles at one-second and one-minute frequency, futures order-book depth, and the published trader-class long/short ratios (top traders by position and by account, and all

accounts); every file is verified against Binance's published checksums. The CFTC Commitments of Traders reports (Traders in Financial Futures and Disaggregated, futures only, through 13 October 2026) come from the CFTC public reporting API. Weekly prices for the CFTC markets (stock index, currencies, Treasury yield and CME futures) come from Yahoo Finance. Processed data and all result files are in the public repository below.

Code availability. All code, pre-specified test specifications, result files and test-unlock counters are public at `https://github.com/Pratikn03/Fianince`. The repository's README lists the scripts in run order, and Supplement S34 maps every number in the paper to the file that produces it.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Use of AI tools. Claude (Anthropic) assisted with data analysis, analysis code, derivations, numerical results and text. Gemini (Google) assisted with planning and internal draft critique. ChatGPT/Codex (OpenAI) assisted with planning, literature discovery, mathematical development, code and Lean proof drafting, executable and kernel checks, result interpretation and manuscript revisions. This assistance was material. The reported kernel checks and computational runs support their explicitly stated results; they do not certify independent human review, novelty or full-manuscript correctness. AI tools are not authors. The human authors remain responsible for verifying all content, citations, analyses and conclusions and for the final scholarly contribution before submission.

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Table 1: What is known and what is new. "Known" rows are used, not claimed.

Result	Status	What the market setting adds, and the closest prior work
Prices identify only a merged response measure; equal-memory groups are inseparable (Theorem 3.1)	Model-specific	The equivalence class of economies with the same public response, made exact for a closed loop; closed-loop identification (Ljung, 1999; Forssell and Ljung, 1999)
Separating K groups takes $(\Delta\eta)^{-(4K-2)}$ experiments; Prony's method attains it (Theorems 6.1, 6.2)	Known rate	Memories are zeros, not poles, of the observed transfer function, so they must first be extracted from the loop (Heinrich and Kahn, 2018; Wu and Yang, 2020; Katz et al., 2023)
General memory: accuracy grows only with $\log N$ (Theorem S17.2)	Known type	Logarithmic rates for exponential mixtures (Rebafka and Roueff, 2015); here for closed-loop price data
With a strategic insider inside the inventory equilibrium, the identification boundary and the rates hold; positive feedback raises impact and makes prices less informative (Theorem 4.6)	**New**	Kyle with risk-averse market makers (Kyle, 1985); new is the joint equilibrium with feedback cohorts and the observer's reduced form
The equilibrium price is the paper's closed loop with effective parameters and the same memory rates, so the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold in equilibrium (Theorem 4.2)	**New**	Removes the reduced-form caveat: the lower-bound pair is realized by two equilibria with identical dealers
Fragility: an equilibrium exists iff dealer risk aversion is below an explicit threshold that falls as $1/G(1)$ (Theorem 4.4)	**New**	Liquidity breakdown with feedback traders; new is the closed-form threshold and its rate
Price impact and the persistence of mispricing are equilibrium outcomes of inventory-averse dealers; feedback raises impact (Theorem 4.1)	**New**	Inventory models of liquidity (Grossman and Miller, 1988; Ho and Stoll, 1981); new is the closed form with exponential-memory feedback, in which the cohorts enter through the Stieltjes transform of the merged measure
With a strategic insider and any causal dealer rule, economies with the same merged measure are outcome-equivalent; group-level quantities are not (Theorem 4.5)	**New**	Links the identification boundary to equilibrium: what prices cannot identify is worth nothing to price-based traders and matters only for targeted actions. Kyle insiders with feedback traders whose rule the dealer knows: Noh (2026)
Manipulation rents from a misjudged aggregate response are first order in the error, which public data	**New**	Manipulation with feedback traders is classical (De Long et al., 1990; Allen and Gale, 1992); new are the link of the rent to the dealer's kernel error, and the contrast with the cost of separat-

Table 2: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

Contract	Channel	Pulses	Benchmark	Saving vs. $T = 20$
I_0: prices, $T = 3$	public prices	271	4,305.25	−2,123.5%
I_0^*: prices, $T = 20$	full price path	12	193.63	0.0%
I_1: prices and signed flow	public flow	12	193.63	0.0%
I_2: noisy equity ($\sigma_E = 0.5$)	statement	12	193.63	0.0%
I_3: exact equity, 5-unit lot	audited equity	1	12.70	93.4%
I_3: exact equity, 0.01 rounding	rounded report	1	15.52	92.0%
I_4: timestamps only	no likelihood	12	193.63	0.0%
I_5: audit registry	internal states	0	0.00	100.0%

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q \delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.